



Osprey "Rest Stops" on the Fall Migration Path

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RESEARCH QUESTION & BACKGROUND

North American ospreys (*Pandion haliaetus*) breed across the Pacific coast, Midwest, and Atlantic coast of North America and spend winter in Central and South America, making round-trip migrations of 5,000-10,000 km.

The intent of this spatial analysis was to identify potential and likely Osprey stop-off locations during their migrations. These locations serve as "rest stops" for the Ospreys during their migration, where the Ospreys rest and refuel for days or weeks. En-route stopover sites are poorly understood relative to breeding and wintering areas, and they are crucial for conservationists to monitor, in addition to winter and summer habitats.

Prior analysis of this dataset (Martell et al. 2004, 2014) documented migration routes and wintering locations through visual inspection of maps. No method for stopover detection was applied, and site overlap between individual birds was not examined.

This study addresses two questions:

1. Is it possible to identify stopover locations from legacy Argos telemetry data?
2. Are there locations that are shared between multiple birds?

DATA

Source: Martell & Douglas (1995–2002), Fall Migration Routes, Timing, and Wintering Sites of North American Ospreys as Determined by Satellite Telemetry. Movebank Data Repository, CC0 Public Domain.

Coverage: 86,383 raw location records reduced to 44,995 valid fixes after filtering to Movebank-verified locations (visible = TRUE). Records span eight years across approximately 75 adult ospreys. Analysis was restricted to the fall migration window (August–December) to align with the original study's intended scope and to avoid conflating fall stopovers with spring movements or winter residency.

Telemetry method: Argos Doppler-shift satellite tracking. Location accuracy varies by quality class: Class 3 (~150 m), Class 1–2 (~350–1,000 m), Class A/B (several kilometers).

Breeding populations: Four regions - Columbia River, OR (West); Minneapolis–St. Paul and Crow Wing County, MN (Midwest); Eastern Long Island, NY and Cape May, NJ (East).

METHODOLOGY

Stopover detection proceeded in four stages:

1. Net Squared Displacement (NSD)
For each point location recorded, the distance from the bird's deployment site was computed. A location was flagged as a stopover candidate if: (a) NSD varied by less than 100 km across a 7-day window, (b) the window spanned at least 3 days, and (c) the bird was at least 100 km from its deployment site. The 100 km threshold accounts for Argos positional error while remaining sensitive to genuine movement.
2. DBSCAN spatial clustering
Flagged locations were grouped together using DBSCAN (Density-Based Spatial Clustering of Applications with Noise; Ester et al. 1996), with a 75 km neighborhood radius and a minimum of 3 locations. The haversine metric was used throughout to ensure correct distances on a sphere.
3. Proximity filtering
Clusters were removed if: (a) mean NSD was below 250 km (indicating the bird was still near its breeding site), or (b) duration exceeded 45 days (indicating winter residency rather than a migration stopover).
4. Shared-site analysis
Distance was computed between each pair of found stopovers from different birds. Pairs within 50 km were classified as using the same site.

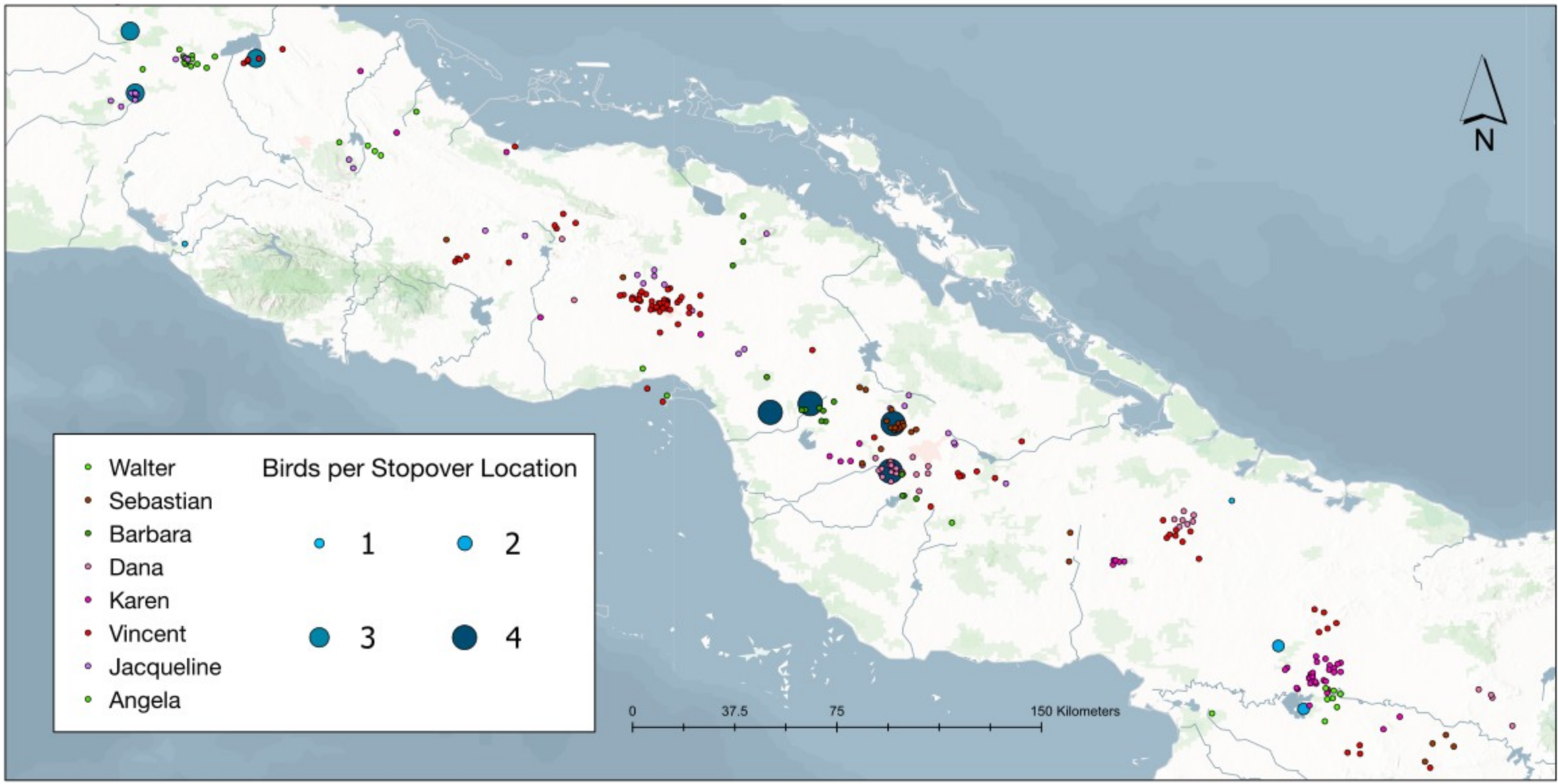


FIGURE A: Cuba (names assigned randomly to birds based on sex)

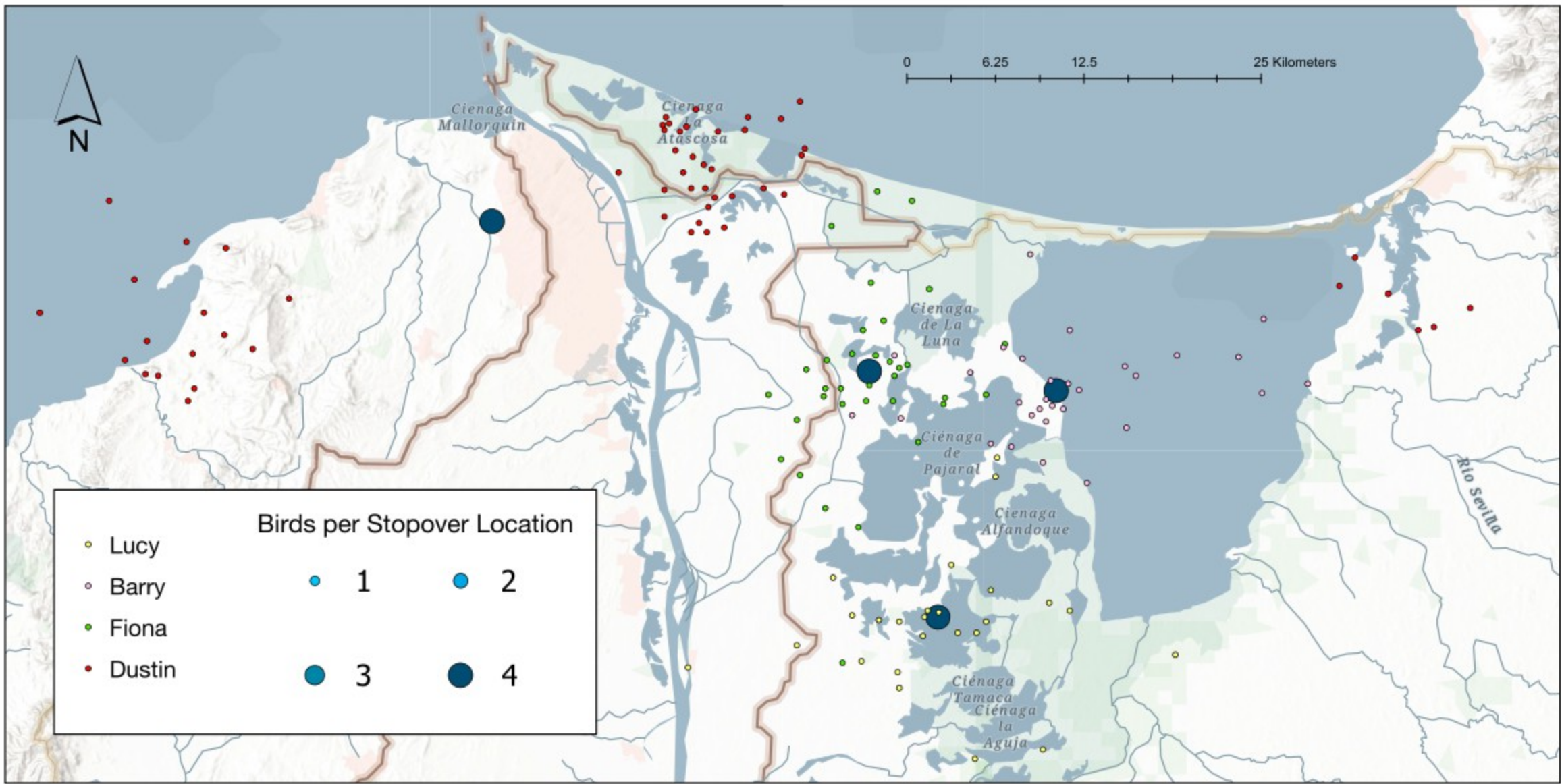


FIGURE B: Colombia (names assigned randomly to birds based on sex)

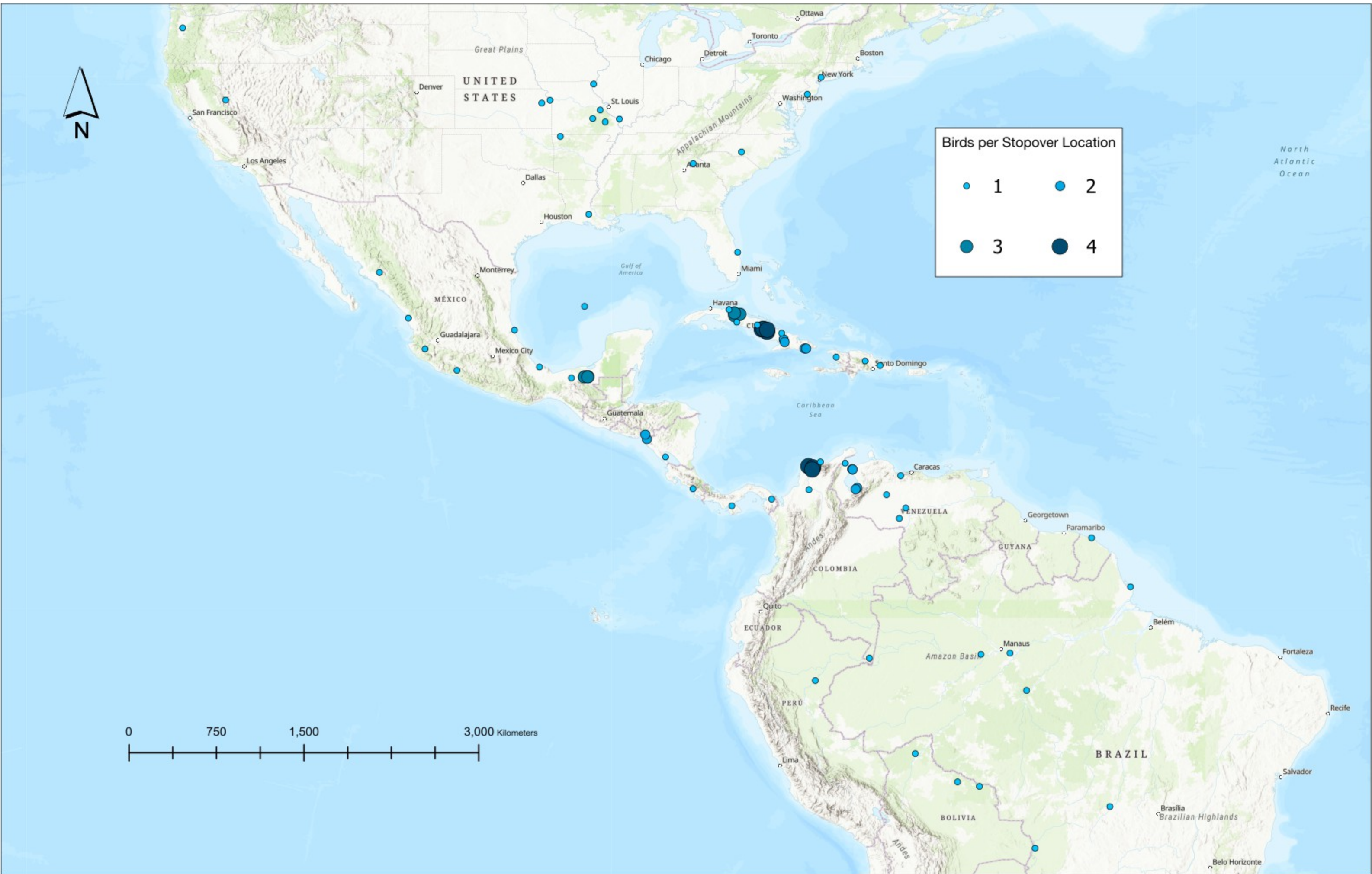
STATISTICAL RESULTS

78 stopovers were identified across 58 birds during fall migration (August–December).

The two most popular sites were a cluster in central Cuba (figure A), used by 4 birds across three separate years (1998–2001), and a cluster along the northern Colombian coast (figure B), used by 4 birds across four years (1995–1999). Both sites drew birds from more than one breeding region.

Duration ranged from 1 to 45 days (the upper limit imposed by the wintering-site filter). Stopovers at higher latitudes (within 1,000 km of breeding sites) were shorter (median 4 days); stopovers in the tropics and at continental barriers were longer (median 12 days).

TOTAL STOPOVERS DETECTED	78
BIRDS WITH 1 OR MORE STOPOVERS DETECTED	58 / 126
MEDIAN STOPOVER DURATION	10 days
MEDIAN POINTS PER STOPOVER	11
STOPOVERS SHARED BY 2 OR MORE BIRDS	24 (31%)
MAXIMUM BIRDS RECORDED AT ONE SITE	4



INTERPRETATION

The two four-bird shared sites correspond to known raptor migration "highways": Cuba sits at the crossing point between the Florida peninsula and the Caribbean island chain, and the northern Colombian coast is a documented area for raptors crossing from Central America to South America. The repeated use of these sites across multiple years and by birds from separate breeding populations suggests these are not incidental stops but functional waypoints shaped by landscape geography.

Thirty-one percent of all detected stopovers were shared by two or more birds. If stopovers were distributed randomly across available habitat, shared-site rates of this magnitude would most likely be improbable given the spatial extent of the study area. The clustering of shared stops at coastlines, island chains, and mountain passes is consistent with habitat-driven site selection rather than chance overlap.

Longer stopovers near continental barriers (Cuba, Colombia) are consistent with the hypothesis that birds pause to rest before traveling over water, where refueling opportunities are limited.

CONCLUSION & IMPLICATIONS

This study demonstrates that the combination of Net Squared Displacement and DBSCAN spatial clustering can reliably identify stopover events from Argos telemetry data.

The identification of central Cuba and the northern Colombian coast as repeatedly used, cross-population stopover sites has direct conservation implications. Both locations fall outside the breeding and wintering areas that have historically received the most attention from conservationists and ecologists.

The limitations for this study include: Argos positional error (Class A/B fixes introduce uncertainty of several kilometers), and small sample size within individual breeding regions that limit population-level comparisons. As well, the original Martell & Douglas study was designed to characterize fall migration routes, timing, and wintering sites. Its intent was not to detect stopovers. The limited amount of data could cause genuine stopovers to go undetected, and the 58 birds with at least one stopover should not be considered comprehensive.

